

Stretch Set — Spatial Olympiad Problems

USAMO / Putnam flavor

For Andy — calibrated to his verified ceiling (IMO/Putnam, solved mentally). June 1, 2026.

How this set differs

The earlier five were format-matched warm-ups (AMC 10–12). These are **ceiling-matched**: each is a real olympiad-grade problem ($\approx 6.5\text{--}8 / 10$), yet still built for Andy's channel — **one spatial insight unlocks a one-line mental computation, and the answer is a single clean value**. No written derivation needed; the work lives as a mental image. Two of them are *invariant* problems (his Windmill sweet spot), and #3 ties straight into the chemistry he already does.

The Five Stretch Problems

1 • Geodesic Around a Cone (unfolding, leveled up)

A cone has slant height **3** and base radius **1**. An ant on the base rim walks **once around the cone** (encircling the apex) and returns to its start. Find the **shortest** such path. $\approx 6.5/10$

- **Insight:** Unroll the cone into a flat sector. Sector radius = slant = 3; arc length = base circumference = 2π , so the sector angle = $2\pi/3 = 120^\circ$. "Once around" = a straight segment from the start point to its copy 120° away.
- Both copies sit at radius 3 from the apex, 120° apart. Chord = $2 \cdot 3 \cdot \sin(60^\circ) = 3\sqrt{3} \approx 5.196$. (Angle $< 180^\circ \Rightarrow$ the straight chord is valid and genuinely loops the apex.)

Answer: $3\sqrt{3}$

For Andy: same "flatten it" move as the spider problem, but on a curved surface — a strictly harder mental unfolding. Pure visualization, no algebra.

2 • The Largest Shadow of a Cube (projection / optimization)

Over all directions of sunlight, what is the **maximum area of the shadow** a **unit cube** casts on a flat floor (orthogonal projection)? $\approx 7.5/10$

- **Insight:** The shadow area onto a plane with unit normal $\mathbf{u}$ equals $\frac{1}{2} \sum |\mathbf{n}_i \cdot \mathbf{u}|$ over the six faces — which simplifies to $|u_x| + |u_y| + |u_z|$.
- Maximize $|u_x| + |u_y| + |u_z|$ subject to $u_x^2 + u_y^2 + u_z^2 = 1 \rightarrow$ max is $\sqrt{3}$, at $\mathbf{u} = (1,1,1)/\sqrt{3}$ (looking straight down the space diagonal). The shadow is then a **regular hexagon**.

Answer: $\sqrt{3}$

For Andy: the surprise is that the biggest shadow comes from the diagonal view and is a hexagon — a spatial-symmetry revelation, recognizable without computation once seen.

3 • The Tetrahedral Angle (3D + chemistry crossover)

From the **center** of a regular tetrahedron, what is the **angle** between the lines to two of its vertices? (This is the methane H–C–H bond angle.) ≈ 6.5/10

- **Insight:** Put the four vertices at alternating cube corners: $(1,1,1)$, $(1,-1,-1)$, $(-1,1,-1)$, $(-1,-1,1)$. The center is the origin.
- For two of them, dot product = $1-1-1 = -1$; each has length $\sqrt{3}$. So $\cos \theta = -1/3 \rightarrow \theta = \arccos(-1/3) \approx 109.47^\circ$.

Answer: $\arccos(-1/3) \approx 109.47^\circ$

For Andy: connects geometry to the molecular structure he already reasons about. The cube-embedding trick is a classic systemizer's shortcut — symmetry replaces trigonometry.

4 • Slicing Space with Planes (invariant / induction — Windmill spirit)

Into how many regions do **n planes in general position** divide 3D space? ≈ 7.5/10

- **Insight:** Each new plane is cut by the previous ones into regions equal to how many parts $n-1$ lines split *that* plane — and each such part slices one region into two. This recursion sums to a clean closed form.
- Result: $R(n) = C(n,0) + C(n,1) + C(n,2) + C(n,3) = (n^3 + 5n + 6)/6$.
- Check: $n=1 \rightarrow 2$, $n=2 \rightarrow 4$, $n=3 \rightarrow 8$, $n=4 \rightarrow 15$. (Note it breaks past 8 — the pattern is *not* "doubles forever," which is the trap.)

Answer: $(n^3 + 5n + 6)/6$

For Andy: a pure invariant/recursion problem — exactly the structure-finding he did on the Windmill and the 2025×2025 grid. The answer is a formula, not a picture, but it's *discovered* by visualizing one new plane at a time.

5 • The Steinmetz Solid (intersection of two cylinders)

Two infinite circular cylinders of radius **r** cross at right angles. Find the **volume of their intersection**. ≈ 8/10

- **Insight:** Slice horizontally at height z . One cylinder forces $|x| \leq \sqrt{r^2 - z^2}$, the other $|y| \leq \sqrt{r^2 - z^2}$. So every cross-section is a **square** of side $2\sqrt{r^2 - z^2}$ — area $4(r^2 - z^2)$.
- Integrate: $\int_{-r}^r 4(r^2 - z^2) dz = 4(2r^3 - \frac{2}{3}r^3) = 16r^3/3$.

Answer: $16r^3/3$

For Andy: the whole problem dissolves the instant you "see" that the slices are squares, not circles — a spatial perception that beats any formula-pushing. The hardest and most beautiful of the five.

Prepared by a3ai for Jacob · Ceiling-matched stretch set. Each rewards a single act of spatial seeing — unfold, project, embed-in-a-cube, add-one-plane, slice — over written manipulation.